

EE345 Mod2-Lec4: Minimum-Norm Solutions

Overview

$$f_{\theta}(x) = \underbrace{x^T}_{\phi(x, x)} \theta$$

Radical basis fns.

1. Module 1: Distances and the Algorithms they Enable

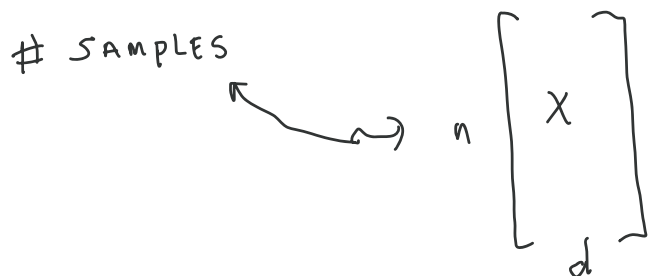
- ▶ Distances and metrics: Euclidean distance, Manhattan distance, cosine distance, worst-case distance, etc.
- ▶ Vectors and matrices: dot product, matrix multiplication, transpose, inverse, etc.
- ▶ Distance based algorithms: k-nearest neighbors (supervised), k-means (unsupervised)

2. Module 2: Linear Models & Regression

- ▶ Convex costs: quadratic functions/forms
- ▶ Linear models and their prevalence in machine learning
- ▶ Least squares: quadratic cost, linear regression
- ▶ **Minimum-norm solutions and introduction to regularization**

3. Module 3: Gradient Descent, Regularization, Kernel Methods

Minimum-Norm Problem



- We saw how to obtain the least squares approximate solution to $X\theta = y$ corresponding to an over-determined set of equations—i.e.,

data matrix $X \in \mathbb{R}^{n \times d}$ with $n \gg d$ ("tall" matrix).

- Here, there is rarely an exact solution to $X\theta = y$, and instead we find an **approximate solution** by minimizing $\|X\theta - y\|^2$.

Underdetermined Systems

An *underdetermined least squares problem* arises when there are fewer equations than unknowns ($n < d$ i.e., X is a "wide" matrix):

$$X = \begin{bmatrix} | & | & \cdots & | \\ x_1 & x_2 & \cdots & x_d \\ | & | & \cdots & | \end{bmatrix}$$

overparameterized
Regime

in which case there are infinitely many solutions to $X\theta = y$.

Intuition: too many features can cause multiple solutions...

find $\hat{\theta}$ s.t. $X\hat{\theta} = y$ and $\|\hat{\theta}\| \leq \|\theta\|$ for any other θ satisfying $X\theta = y$

Minimum-Norm Problem

$$\min_{\theta} \|\theta\|_2^2 \text{ subject to } X\theta = y.$$

(min-norm-problem)

$$\min_{\theta \in \Theta} \mathcal{L}(\theta)$$

$$\theta \in \Theta$$

$$\Theta = \{\theta \in \mathbb{R}^d \mid X\theta = y\}$$

constraint set.

$$\begin{matrix} n & \left[\begin{array}{c} \sim \\ \sim \end{array} \right. & X & \left. \begin{array}{c} \sim \\ \sim \end{array} \right] \\ & & d & \end{matrix}$$

When does this problem arise and why do we care?

From Classical to Modern ML

- **Classical view:** Too few parameters $\Rightarrow$ *underfitting*.
- **Modern reality:** *Overparameterization* occurs in models with **more parameters than data**.
- Deep nets with millions of weights can *interpolate* (zero training error) yet still *generalize*.
- Key question: *Of the many solutions, which one is "stable" (i.e. robust to small changes in the data)? And, What selection method produces a "good" fit?*

Real-World Examples of Underdetermined Systems

Deep Neural Networks.

- ResNet-50 has about 25 million parameters, while a medical imaging dataset may have only a few thousand training examples.
- Despite $d \gg n$, gradient-based training often achieves zero training error.
- The solution behaves as if it were a *minimum-norm* solution in function space, illustrating the **implicit regularization** of **stochastic gradient descent (SGD)**.

Learning word embeddings or language models.

- In word2vec or transformer pretraining, the number of parameters in the embedding or attention layers greatly exceeds the number of distinct word co-occurrences observed per batch.
- Weight decay or **early stopping** biases the solution toward low-norm parameter matrices.

Real-World Examples of Underdetermined Systems

Genomics and Biomedical Prediction. Consider predicting disease risk from gene expression.

- A typical dataset may have $n = 100$ patients but $d = 20\text{k}$ gene features.
- **Ridge regression** makes the problem well-posed:

$$\min_{\theta} \|X\theta - y\|_2^2 + \lambda\|\theta\|_2^2,$$

yielding a minimum-norm solution that stabilizes predictions.

- This also occurs in *bag-of-words* text classification tasks.

Real-World Examples of Underdetermined Systems

Redundant robot arm (inverse kinematics).

- A 7-DOF robotic arm can position its end-effector in 3D, creating redundancy:

$$J(\theta)\dot{\theta} = \dot{x}, \quad J \in \mathbb{R}^{3 \times 7}.$$

- The minimum-norm velocity

$$\dot{\theta} = J^{\top} (J J^{\top})^{-1} \dot{x}$$

achieves the desired motion with minimal total joint movement, leading to smoother, energy-efficient trajectories.

Learning a robot control policy from demonstrations.

- A neural policy network may contain millions of parameters but only thousands of example trajectories.
- Weight decay or early stopping biases learning toward a minimum-energy control policy that matches the expert behavior without overfitting.

Problem: Finding the Minimum Norm Solution

$$X\theta = y$$

KKT

Optimality conditions
for constrained
opt.

We want to find the smallest-norm vector θ that satisfies the linear system

$$X\theta = y,$$

where $X \in \mathbb{R}^{n \times d}$ has full row rank ($n < d$) and the system is consistent. This problem is underdetermined and can be cast as the following optimization problem:

$$\min_{\theta} \|\theta\|^2 \quad \text{s.t. } X\theta = y.$$

$$\mathcal{J}(\theta, \lambda) = \|\theta\|^2 + \lambda^T (X\theta - y)$$

Problem: Finding the Minimum Norm Solution

Part (a): Instead of solving the constrained problem, consider the augmented objective

$$J(\theta) = \frac{1}{2} \|\theta\|_2^2 + \lambda^\top (y - X\theta), \quad \text{where } \lambda \in \mathbb{R}^m \text{ is a fixed vector.}$$

Compute $\nabla_\theta J(\theta)$ and find θ that minimizes $J(\theta)$ in terms of λ . $\theta(\lambda)$

$$J(\theta) = \frac{1}{2} \theta^\top \theta + \lambda^\top (y - X\theta)$$

$$\nabla_\theta J(\theta) = \theta - X^\top \lambda = 0 \quad \Rightarrow$$

$$\theta(\lambda) = X^\top \lambda$$

Problem: Finding the Minimum Norm Solution

Problem: Finding the Minimum Norm Solution

Part (b): Substitute the result into $X\theta = y$ and solve for λ in terms of X and y .

$$\underline{\theta(\lambda) = X^T X}$$

Plugging it in we get $X\theta(\lambda) = y \Leftrightarrow \underbrace{XX^T}_{n \times n} \lambda = y \Rightarrow \boxed{\lambda = (XX^T)^{-1} y}$

$X^T X \rightarrow$ full col rank
rank d or d L.I. cols.

$$d \begin{bmatrix} X^T \\ n \end{bmatrix} \begin{bmatrix} \\ d \end{bmatrix} \rightarrow X^T X \quad d \times d$$

XX^T is invertible

$$\begin{matrix} n \\ \uparrow \\ \left[\begin{array}{c} X \\ d \end{array} \right] \end{matrix} \begin{matrix} \\ d \\ \uparrow \\ \left[\begin{array}{c} X^T \\ n \end{array} \right] \end{matrix} \rightarrow n \times n$$

full row rank $\Leftrightarrow \text{rank}(X) = n$
 $\Leftrightarrow n$ L.I. rows

Problem: Finding the Minimum Norm Solution

Part (c): Substitute λ back to get an explicit expression for θ . Show that

$$\hat{\theta} = X^T (X X^T)^{-1} y.$$

$$\hat{\theta} := \theta(\lambda) = X^T \lambda = X^T (X X^T)^{-1} y$$

$$\underline{J(\theta) = \frac{1}{2} \|\theta\|^2 + \lambda^T (y - X\theta)}$$

Problem: Finding the Minimum Norm Solution

Part (d): Explain why this $\hat{\theta}$ is the same as the solution to

$$\min_{\theta} \|\theta\|_2 \quad \text{s.t. } \underline{X\theta = y.}$$

$$\hat{\theta} = X^T (X X^T)^{-1} y.$$

$S := \{\theta \in \mathbb{R}^d \mid X\theta = y\}$. Pick arbitrary $\theta \in S$. so that $X\theta = y$

thus $X(\theta - \hat{\theta}) = 0$. Now, claim that $\boxed{\theta - \hat{\theta} \perp \hat{\theta}}$

Indeed $\underline{(\theta - \hat{\theta})^T \hat{\theta} = 0}$ since $\underline{\text{range}(X^T) \perp \text{null}(X)} \Rightarrow \underline{\text{FROL (Lin. Alg Fact)}}$

Next step.

$$\|\hat{\theta}\|^2 \leq \|\theta\|^2 \quad \text{for any other } \theta \in S$$

Problem: Finding the Minimum Norm Solution

$$\left[(\theta - \hat{\theta})^T \hat{\theta} = \underbrace{(\theta - \hat{\theta})^T X^T (X X^T)^{-1} y}_{=0} = \underbrace{(X(\theta - \hat{\theta}))^T}_{=0} \underbrace{(X X^T)^{-1} y}_{\text{since } \theta, \hat{\theta} \in S = \{\theta' \mid X\theta' = y\}} = 0 \right]$$

$$\underbrace{\|\theta\|^2}_{=} = \|\hat{\theta} + \theta - \hat{\theta}\|^2 = (\hat{\theta} + \theta - \hat{\theta})^T (\hat{\theta} + \theta - \hat{\theta}) = \|\hat{\theta}\|^2 + 2 \underbrace{(\theta - \hat{\theta})^T \hat{\theta}}_{=0} + \|\theta - \hat{\theta}\|^2$$

$$\underbrace{\|\theta\|^2}_{=} = \|\hat{\theta}\|^2 + \underbrace{\|\theta - \hat{\theta}\|^2}_{\geq 0} \geq \underbrace{\|\hat{\theta}\|^2}_{=}$$

Problem: Finding the Minimum Norm Solution

$$\min \{ \|\theta\|^2 \text{ s.t. } X\theta = y \}$$

• "dual" / Lagrange variables $\Rightarrow J(\theta, \lambda)$ aug. cost

• Opt. w.r.t. λ, θ to get $\hat{\theta}_{mn} = X^T (X X^T)^{-1} y$

$$\theta_{LS} = \underbrace{(X^T X)^{-1}} X^T y$$

$X^T X$ not invertible

$$X X^T$$

Problem: Finding the Minimum Norm Solution

Part (e): Let

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Find the minimum norm solution to $X\theta = y$.

Problem: Finding the Minimum Norm Solution

• min norm soln is "stable"

So, why do we choose the minimum norm solution then? Coming back to the key questions at the top...

$$f(x) = y$$

$$f(x + \delta x) = y + \delta y$$

$$\|x\theta - y\|^2 + \lambda R(\theta)$$

Setup: Underdetermined Least Squares

- Suppose $X \in \mathbb{R}^{n \times d}$ with $n < d$ and $\text{rank}(X) = n$.

$$\frac{1}{(X X^T)^{-1}}$$

$$n \begin{bmatrix} \text{---} x_1^T \text{---} \\ \vdots \\ \text{---} x_n^T \text{---} \end{bmatrix}$$

$$X\theta = y$$

$$\begin{array}{ll} \min & \| \theta \|^2 \\ \text{s.t.} & X\theta = y \end{array} \leftarrow$$

$$\left[\cdot \right] \quad \{ \theta : X\theta = y \} = \underline{\theta_p + \mathcal{N}(X)}$$

$$X\theta_p = y, \quad \tilde{\theta} \in \mathcal{N}(X)$$

$$X(\theta_p + \tilde{\theta}) = y$$

$$X\theta_p + \cancel{X\tilde{\theta}} = y$$

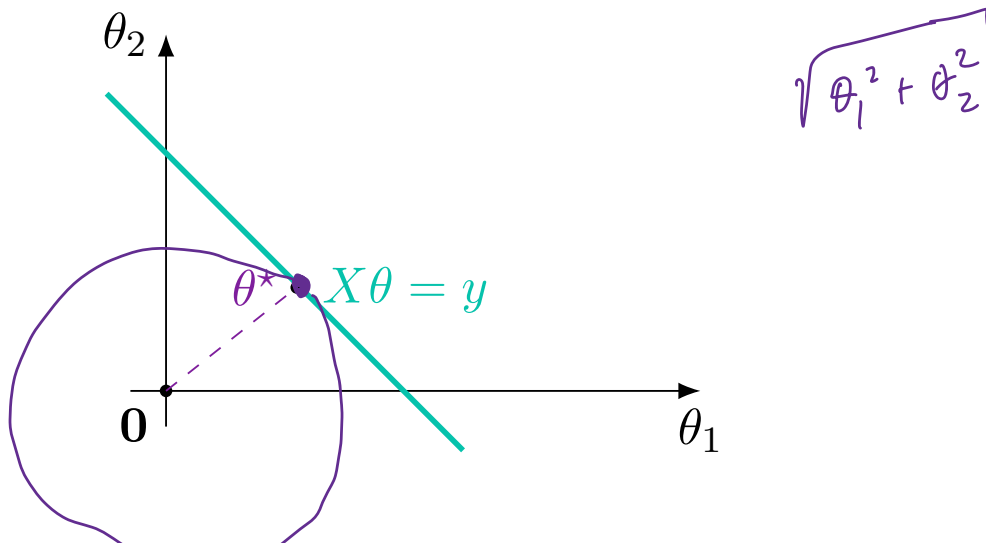
The Minimum-Norm Interpolant

- Define

$$\theta^* = \arg \min_{\theta} \|\theta\|_2^2 \quad \text{s.t.} \quad X\theta = y.$$

$$\theta^* = X^T (X X^T)^{-1} y = X^\dagger y$$

- Geometric view:



Why Min-Norm: Because of the Robustness

- Suppose y changes slightly: $y \mapsto y + \delta y$.

$$\theta^*(y) = X^T (X X^T)^{-1} y$$

$$\begin{bmatrix} 45+2 \\ 54+3 \\ 60-4 \end{bmatrix}$$

$$\theta^*(y + \delta y) = X^T (X X^T)^{-1} (y + \delta y) = \theta^*(y) + X^T (X X^T)^{-1} \delta y$$

$$\left\{ \|\theta^*(y + \delta y) - \theta^*(y)\| \leq \|X^T (X X^T)^{-1}\| \|\delta y\| \right\}$$

$$f(x) = A x$$

$y \mapsto \theta^*(y)$ is linear ; "Lipschitz continuous"

i.e. small changes in the data, we small parameter changes θ

$$X\theta = y$$

$$\begin{aligned} X\theta &= y \\ \parallel \theta \parallel^2 &\leftarrow \end{aligned}$$

"measure how good your sensor is"

Why Other Solutions Are Less Stable

- Any other interpolant can be written as

$$\theta = \theta^* + v, \quad v \in \mathcal{N}(X).$$

$$X\theta = y$$

$$\theta^* = X^T(XX^T)^{-1}y$$

$$\| \theta \|^2$$

- v does not affect $X\theta = y$, but can have large magnitude.
- If y changes slightly, the new affine set $\{\theta : X\theta = y + \delta y\}$ shifts.
- So if you have a data-dependent (i.e., dependent upon $(X, y + \delta y)$) selection rule for picking solutions then $v(\delta y)$ could vary wildly.
- The **minimum-norm** solution means *no nullspace component* $\implies$ most stable fit.
- For predictions and for perturbations in X : carrying extra nullspace mass is undesirable; minimum-norm avoids it and provides a clean inductive bias.

Summary: Robustness of the Minimum-Norm Solution

- All exact fits satisfy $X\theta = y$, but they differ in sensitivity to data.
- The minimum-norm solution

$$\theta^* = X^\top (XX^\top)^{-1}y$$

is **most robust**:

- ▶ It changes smoothly with y .
 - ▶ It contains no nullspace component that can amplify noise.
 - ▶ It provides the “simplest” exact fit.
- This principle underlies regularization and implicit bias in modern ML: *prefer simple (low-norm), stable solutions*.

From Stability to Regularization

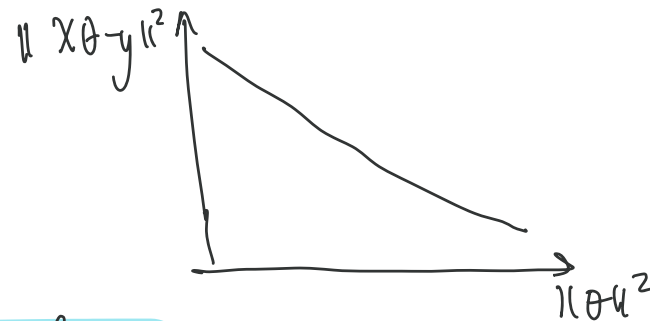
- The minimum-norm solution is a **stable** interpolant: → fitted model

$$\theta^* = \arg \min_{\theta} \|\theta\|_2^2 \quad \text{s.t.} \quad X\theta = y.$$

$$\left[\begin{array}{l} \text{fitted model} \\ \text{penalty} \end{array} \right] \rightarrow \|\theta\|^2 + \frac{\lambda}{2} (y - X\theta)^2$$

- In practice, data are noisy — exact interpolation can overfit.
- General Idea: **trade off fit and simplicity via regularization.**

$$\min_{\theta} \|X\theta - y\|^2 + \lambda \|\theta\|_2^2$$



• $\lambda > 0$ control trade off.

• As $\lambda \rightarrow 0$, we recover the minimum norm solution

• As $\lambda \rightarrow \infty$, we prioritize simplicity ($\theta \approx 0$)

Proof of Minimum-Norm Equivalence

$$\min_{\theta} \|X\theta - y\|^2 + \lambda \|\theta\|^2 \quad \swarrow$$

$$\min_{\theta} \left\| \begin{bmatrix} X \\ \lambda^{1/2} I \end{bmatrix} \theta - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|^2$$

$$\begin{bmatrix} X \\ \sqrt{\lambda} \begin{bmatrix} 0 & \ddots & 0 \\ & \ddots & \\ 0 & & \sqrt{\lambda} \end{bmatrix} \end{bmatrix}$$

$$\min_{\theta} \|\tilde{X} \theta - \tilde{y}\|^2 \rightarrow \theta = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \tilde{y}$$

$$\theta = \left(\begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^T \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \right)^{-1} \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^T \begin{bmatrix} y \\ 0 \end{bmatrix} = \underbrace{(X^T X + \lambda I)^{-1}}_{\lambda \rightarrow 0} X^T y$$

$$\underbrace{X^T X} + \underbrace{\lambda I}$$

$\lambda \rightarrow 0$

Proof of Minimum-Norm Equivalence

$$\theta = \underbrace{(X^T X + \lambda I)^{-1}}_{\text{?}} X^T y \quad \left[\begin{array}{c} \text{?} \\ \text{?} \end{array} \right]$$

• Can't do $\lambda \rightarrow 0$, $X^T X$ is not invertible

• FACT: $X^T X X^T + \lambda X^T$ can be factored out as follows

$$\left[\begin{array}{l} (X^T X + \lambda I) X^T = \underbrace{X^T X X^T + \lambda X^T}_{\text{?}} = \underbrace{X^T (X X^T + \lambda I)}_{\text{?}} \end{array} \right]$$

$$(X^T X + \lambda I) X^T = X^T (X X^T + \lambda I)$$

$$\underbrace{X^T (X X^T + \lambda I)^{-1}}_{\text{?}} = (X^T X + \lambda I)^{-1} X^T$$

Proof of Minimum-Norm Equivalence (The Tricky Part)

$$\theta = (X^T X + \lambda I)^{-1} X^T y$$

$$\theta = \underbrace{X^T (X X^T + \lambda I)^{-1}}_{\text{is invertible}} y$$

now, $\lambda \rightarrow 0$ $\theta_{\lambda \rightarrow 0} = \underbrace{X^T (X X^T)^{-1}}_{\text{is invertible}} y$ ★

$\min_{\theta} \|X\theta - y\|^2 + \cancel{\lambda \|\theta\|^2}$

$n \ll d$

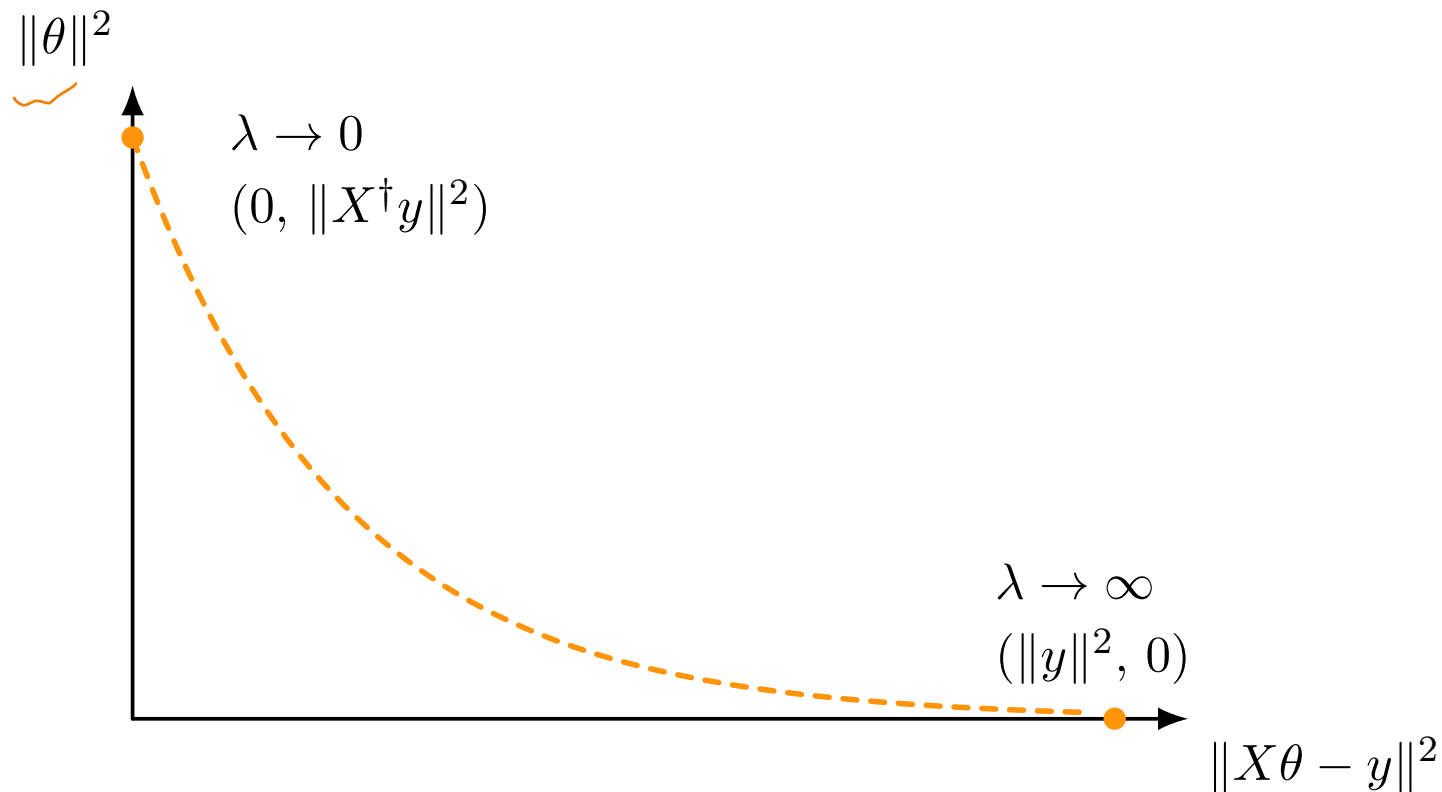
$$\underbrace{\begin{bmatrix} X \\ \vdots \end{bmatrix}}_{n \times n} \begin{bmatrix} X^T \\ \vdots \end{bmatrix}$$

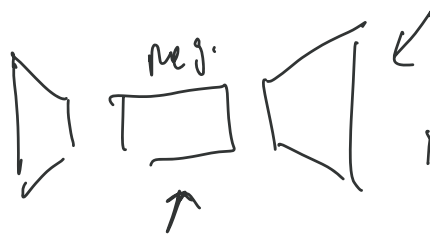
$$\begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix}$$

Proof of Minimum-Norm Equivalence (The Tricky Part)

Tradeoff Visualization

$$\min_{\theta} \underbrace{\|X\theta - y\|_2^2}_{\text{data fit}} + \lambda \underbrace{\|\theta\|_2^2}_{\text{regularization}}$$





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Regularization

VMLS

Regularization: Additional penalty term added to the cost function to encourage a solution with desirable properties.

Regularized Least Squares:

$$\min_{\theta} \|X\theta - y\|_2^2 + \lambda R(\theta)$$

- $R(\theta)$ is the regularization term.
- λ is the regularization parameter.
- The model has different names depending on $R(x)$ (e.g., ridge regression, LASSO, etc.)

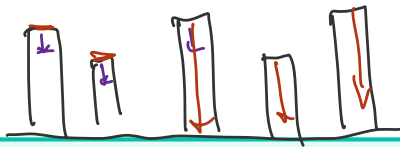
Sparse soln.

$$\begin{bmatrix} \uparrow & \uparrow & & & \\ \theta_1 & \theta_2 & \dots & \dots & \theta_d \end{bmatrix}$$

↑

↑
 d

System Id. ↑
Compressed sensing

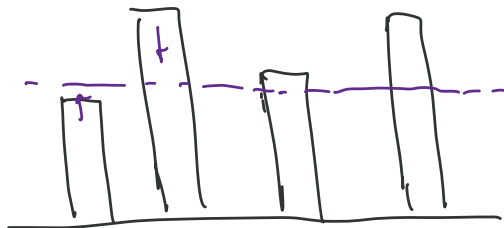


Regularization

$$X\theta = y$$

$$\min_{\theta} \|X\theta - y\|_2^2 + \lambda R(\theta)$$

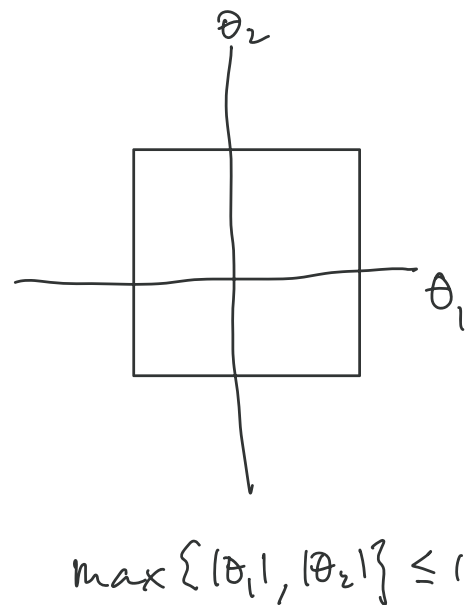
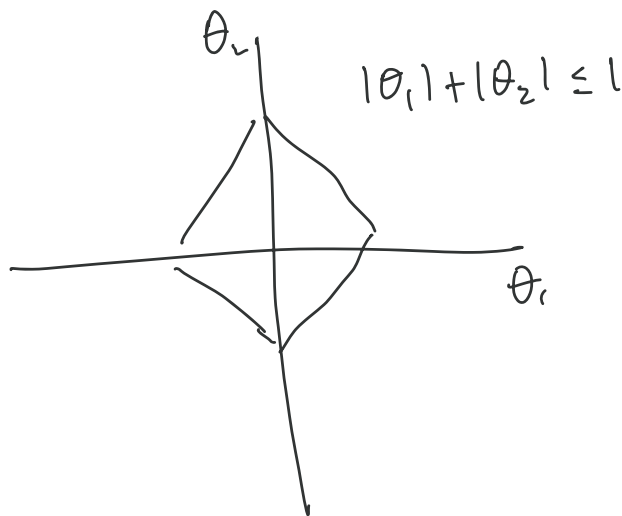
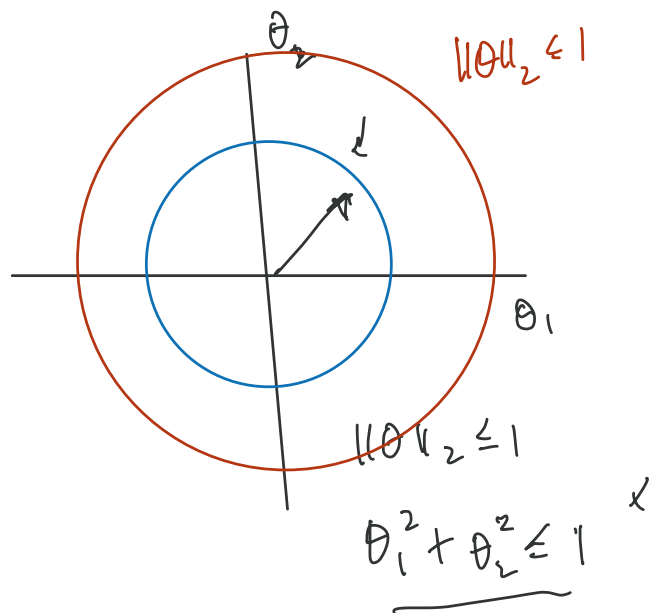
- If $R(\theta) = \|\theta\|_2^2 = \theta_1^2 + \theta_2^2 + \dots + \theta_d^2$. ℓ_2 regularization, Tikhonov reg.
Ridge regression
it effectively smooths the soln.
- If $R(\theta) = \|\theta\|_1 = |\theta_1| + |\theta_2| + \dots + |\theta_d|$. ℓ_1 reg. or LASSO
it sparsifies
- If $R(\theta) = \|\theta\|_\infty = \max\{|\theta_1|, |\theta_2|, \dots, |\theta_d|\}$
it equalizes the soln.



Norm Balls

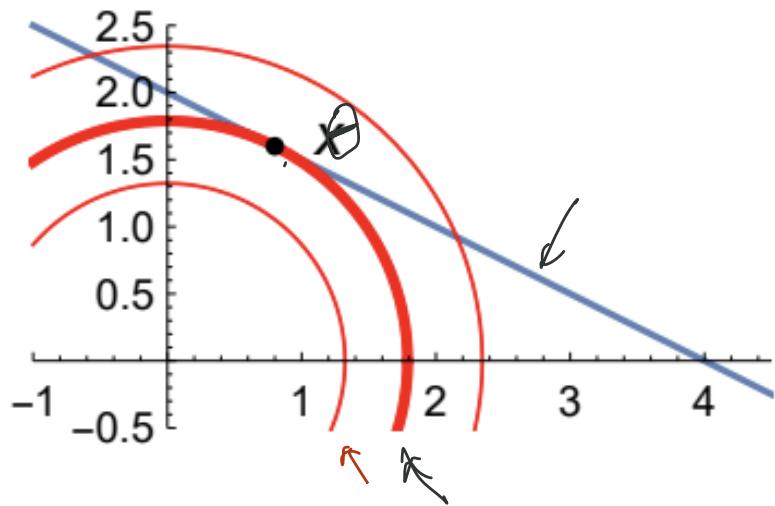
For a norm $\|\theta\|_p$, then define a normball of radius r :

$$B_r = \{\theta \in \mathbb{R}^d \mid \|\theta\|_p \leq r\} \quad p \in \{1, 2, \infty\}$$



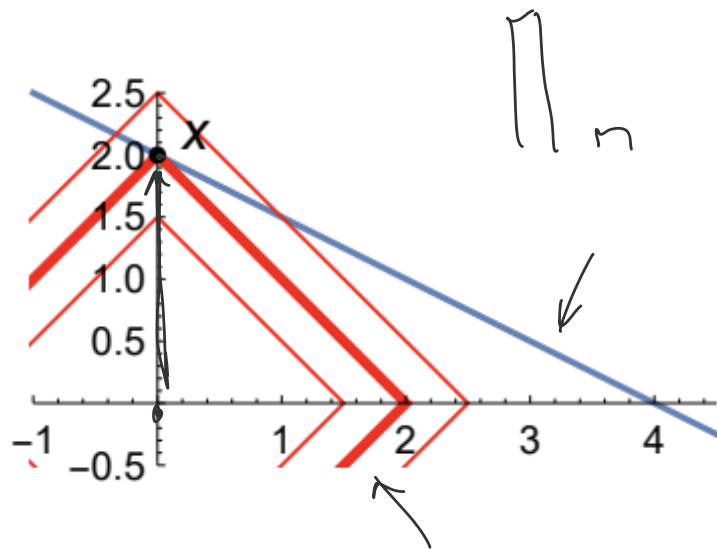
Simple Example

$$\min_{\theta} \{ \|\theta\|_p \mid X\theta = y \}$$



- set of solns is affine space $\{ \theta \mid \underbrace{X\theta = y} \}$
- soln. is a point belongs smallest norm ball intersecting $X\theta = y$

Simple Example

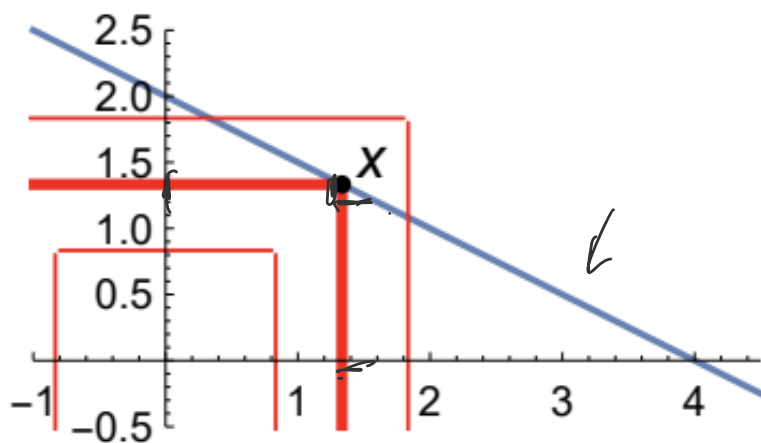


$$\|\theta\|_1 \leq 2$$

(stochastic) grad.

$$\|x\theta - y\|^2 + \lambda \|\theta\|_1$$

$$X\theta = y$$



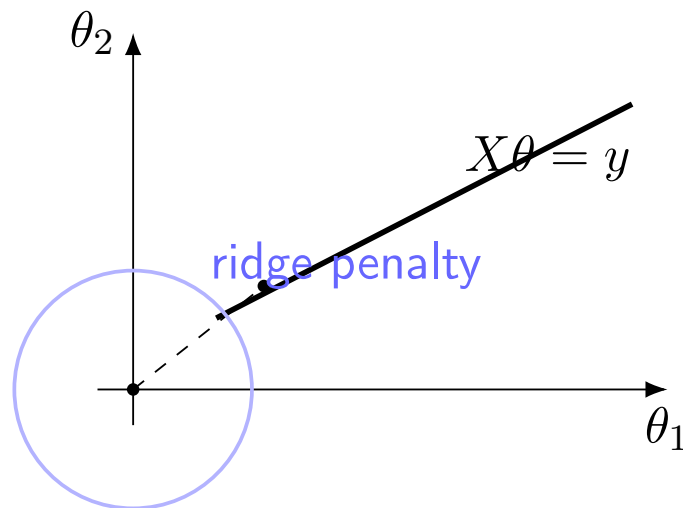
$$\min \|\theta\|_p$$

$$\text{s.t. } X\theta = y$$

$$p \in \{1, 2, \infty\}$$

Regularization = Controlled Stability

- Regularization keeps the solution **stable** when y changes slightly.
- It also reduces **variance** and improves **prediction quality**.
- Gradient descent with small initialization often behaves as if λ is small but positive (implicit regularization).



Geometrically: regularization picks the point on the affine set $X\theta \approx y$ closest to the smallest-norm ball.

Why Regularization Improves Prediction

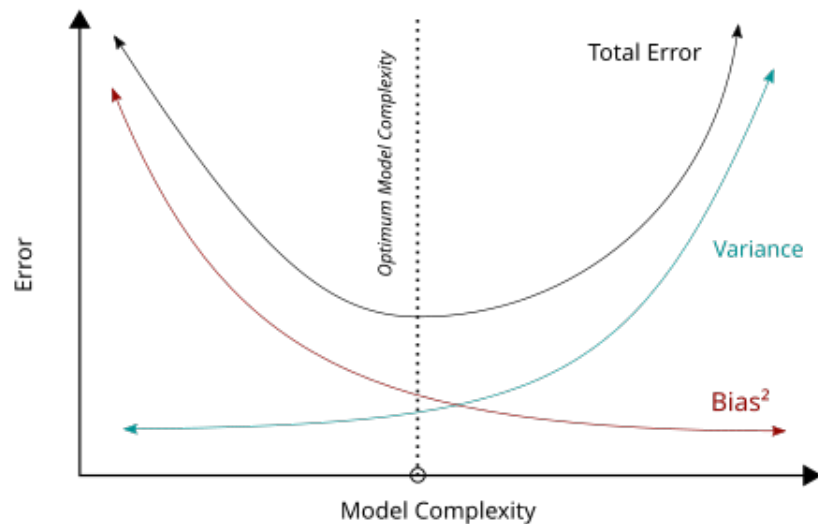
- Model: $y = X\theta^* + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$
- Ridge solution:

$$\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

- Bias–variance decomposition:

$$\mathbb{E}\|\hat{y} - X\theta^*\|^2 = \underbrace{\|X(\mathbb{E}[\hat{\theta}_\lambda] - \theta^*)\|^2}_{\text{bias}^2} + \underbrace{\text{tr}(X \text{Var}[\hat{\theta}_\lambda] X^\top)}_{\text{variance}} + \text{noise}.$$

- Increasing λ :
 - ▶ $\uparrow$ bias (shrink coefficients)
 - ▶ $\downarrow$ variance (less noise amplification)
- Optimal λ minimizes total error $\Rightarrow$ better generalization.



Gradient Descent Dynamics in Least Squares

- Objective:

$$f(\theta) = \frac{1}{2} \|X\theta - y\|^2, \quad \nabla f(\theta) = X^\top (X\theta - y).$$

Unrolling the Gradient Descent Recursion

Unrolling the Gradient Descent Recursion

Using the Matrix Geometric Series

$$\theta_t = \eta \sum_{k=0}^{t-1} (I - \eta X^\top X)^k X^\top y.$$

- Let $B = X^\top X$. The finite geometric series gives

$$\sum_{k=0}^{t-1} (I - \eta B)^k = (\eta B)^{-1} [I - (I - \eta B)^t].$$

Finite Steps $\implies$ Implicit Regularization

- For a finite number of steps t ,

$$\theta_t = (X^\top X)^{-1} [I - (I - \eta X^\top X)^t] X^\top y.$$

- For small $\eta X^\top X$,

$$(I - \eta X^\top X)^t \approx (I + \eta t X^\top X)^{-1}.$$

- Hence approximately:

$$\boxed{\theta_t \approx \left(X^\top X + \frac{1}{\eta t} I \right)^{-1} X^\top y.}$$

- This has the same form as **ridge regression**:

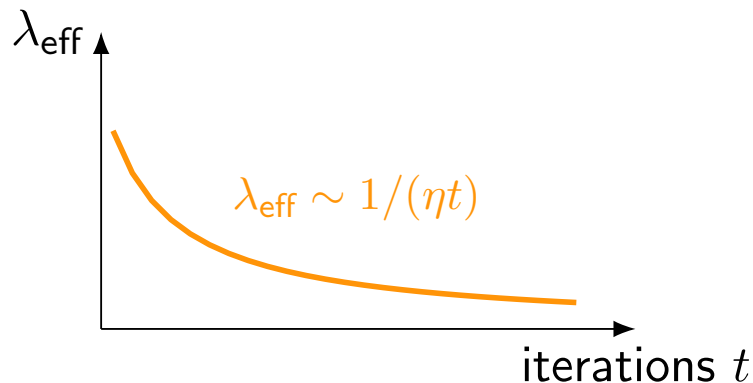
$$\theta_\lambda = (X^\top X + \lambda I)^{-1} X^\top y, \quad \text{with } \lambda_{\text{eff}} \approx \frac{1}{\eta t}.$$

Interpretation: “Implicit Regularization”

- Even though we did *not* add an explicit penalty, gradient descent acts as if there were a small ridge term.
- **Early stopping** corresponds to a larger effective λ : smoother, smaller-norm solution.
- **Running to convergence** corresponds to $\lambda_{\text{eff}} \rightarrow 0$: recovers the minimum-norm interpolant.

Gradient descent with small initialization behaves like ridge regression with $\lambda_{\text{eff}} \approx \frac{1}{\eta t}$.

Stopping gradient descent early acts like adding an ℓ_2 penalty: controls model complexity without explicit regularization.



more training $\implies$ less regularization

From Linear Models to Kernels

- For linear regression, “simple” means small $\|\theta\|$.
- For nonlinear functions, we need a notion of “smoothness” or “simplicity” in function space.
- In a Reproducing Kernel Hilbert Space (RKHS):

$$\min_{f \in \mathcal{H}} \sum_{i=1}^m (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}}^2.$$

- The solution (by the Representer Theorem) is

$$f_{\lambda}(x) = k(x)^{\top} (K + \lambda I)^{-1} y, \quad K_{ij} = k(x_i, x_j).$$

- Same principle:
 - ▶ Fit the data (small residuals),
 - ▶ Keep the function “small” in $\mathcal{H}$ (smooth, stable).

Unifying Principle

Minimum-Norm and Regularization

Everywhere we've looked:

Linear least squares: $\min_{\theta} \|X\theta - y\|^2 + \lambda\|\theta\|^2,$

Kernel regression: $\min_{f \in \mathcal{H}} \sum_i (f(x_i) - y_i)^2 + \lambda\|f\|_{\mathcal{H}}^2.$

- Both choose the **most stable, simplest** fit consistent with the data.
- λ tunes how much we trust data vs how much we smooth.
- As $\lambda \downarrow 0$, we recover the minimum-norm interpolant.

The core idea: prefer simple (low-norm, smooth) solutions—they're the most stable and the most likely to generalize.